

Products, projections and orientation

RECALL

Dot product and its sign

$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$. For nonzero inputs this also equals $|\mathbf{a}| |\mathbf{b}| \cos\theta$. Positive, zero and negative distinguish acute, right and obtuse tail-to-tail angles. $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.

A dot product is a scalar. The component rule also allows zero inputs, although their angle is undefined.

Angle from components

$$\theta = \cos^{-1}\left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}\right).$$

Both vectors must be nonzero; use the angle between 0° and 180° . Keep the numerator's sign. Check the calculator's angle mode.

Signed component and vector projection

For $\mathbf{b} \neq 0$,

$$s = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|}, \quad \mathbf{p} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}} \mathbf{b}.$$

The scalar s uses the direction of $\mathbf{b}$; $\mathbf{p}$ is the vector on its line. $|\mathbf{p}| = |s|$.

Both have the units of $\mathbf{a}$. A signed component may be negative; its magnitude cannot. Do not substitute one output type for the other.

Cross-product meaning

In three dimensions, $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ is a vector perpendicular to both inputs. For nonzero inputs,

$$|\mathbf{c}| = |\mathbf{a}| |\mathbf{b}| \sin\theta.$$

Order fixes direction by the right-hand rule.

Use a right-handed frame. $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$. A zero result has no normal direction.

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UNDERSTAND

Why the projection multiplier works

Write $p = cb$ and require $(a - cb) \cdot b = 0$. This gives $c = (a \cdot b) / (b \cdot b)$. The remainder is $q = a - p$.

For nonzero b the denominator is positive, so the multiplier is unique. Reconstruction alone does not prove that q is perpendicular.

Checks with different purposes

For a projection, check p lies on b 's line and $q \cdot b = 0$. Also check $p + q = a$. Then $|a|^2 = |p|^2 + |q|^2$. For a cross product, check perpendicularity, scale and ordered orientation separately.

Both signs of a normal pass perpendicularity checks. Numerical tests support selected cases; they do not prove a general identity.

Zero and reversal

For nonzero inputs: dot zero means perpendicular; cross zero means parallel, including opposite directions. Replacing b by any nonzero multiple leaves the vector projection unchanged. A negative multiple reverses the signed component's sign.

Zero input a onto nonzero b gives zero projection. Zero reference b makes projection undefined. Never infer an angle from a zero input.

Area, normal and perpendicular distance

For adjacent edge vectors u and v , parallelogram area is $|u \times v|$; triangle area is half this. For a nonzero cross product, $\hat{n} = (u \times v) / |u \times v|$. The distance of a 's tip to b 's line through the origin is $|q|$.

Form edges from a common vertex. Length-edge products have area units; the unit normal is dimensionless. Swapping edges reverses the normal, not the area.

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CONSULT

Ordered component rule

For $\mathbf{c} = \mathbf{a} \times \mathbf{b}$,

$$c_x = a_y b_z - a_z b_y,$$

$$c_y = a_z b_x - a_x b_z,$$

$$c_z = a_x b_y - a_y b_x.$$

The cyclic order matters. For inputs in the xy -plane only c_z can be nonzero, and the magnitude is $|c_z|$. Its sign records direction along the z -axis.

Supplied physical models

Constant-force work: $W = \mathbf{F} \cdot \mathbf{d}$, with $\mathbf{d}$ the displacement of the force's application point. Torque about O : $\boldsymbol{\tau}_O = \mathbf{r} \times \mathbf{F}$, with $\mathbf{r}$ from O to that point. Its magnitude is $|\mathbf{F}| \ell$, where ℓ is the perpendicular lever arm.

Work model requires force constant in magnitude and direction. Torque must name the reference. Use J for work and N m for torque. Neither one force's work nor its torque alone determines the motion.

Units and input meanings

A product carries the product of its inputs' units. An angle ratio is dimensionless. A projection has the units of the vector projected, independent of the reference vector's scale. Position, displacement and perpendicular distance have different roles even when all are measured in metres.

Check both units and the quantity requested. Correct units cannot distinguish work from torque or a position from a displacement.

The small projection program

Use three numeric entries in x, y, z order; calculate bb , reject zero before division, calculate ab , then return each entry of $\mathbf{b}$ multiplied by ab/bb . Return None for a zero reference, not a zero vector.

Compare ordinary, perpendicular, parallel and zero cases with hand predictions.

P101 notebook/list/condition/function readiness first. Our code assumes modest finite numeric inputs; it is not a general numerical library. Rounding discrepancies require interpretation, not automatic removal.